

# Leave-one-out disturbance ablation

Train on everything, then drop one disturbance at a time and see what the success rate does.

## What was run

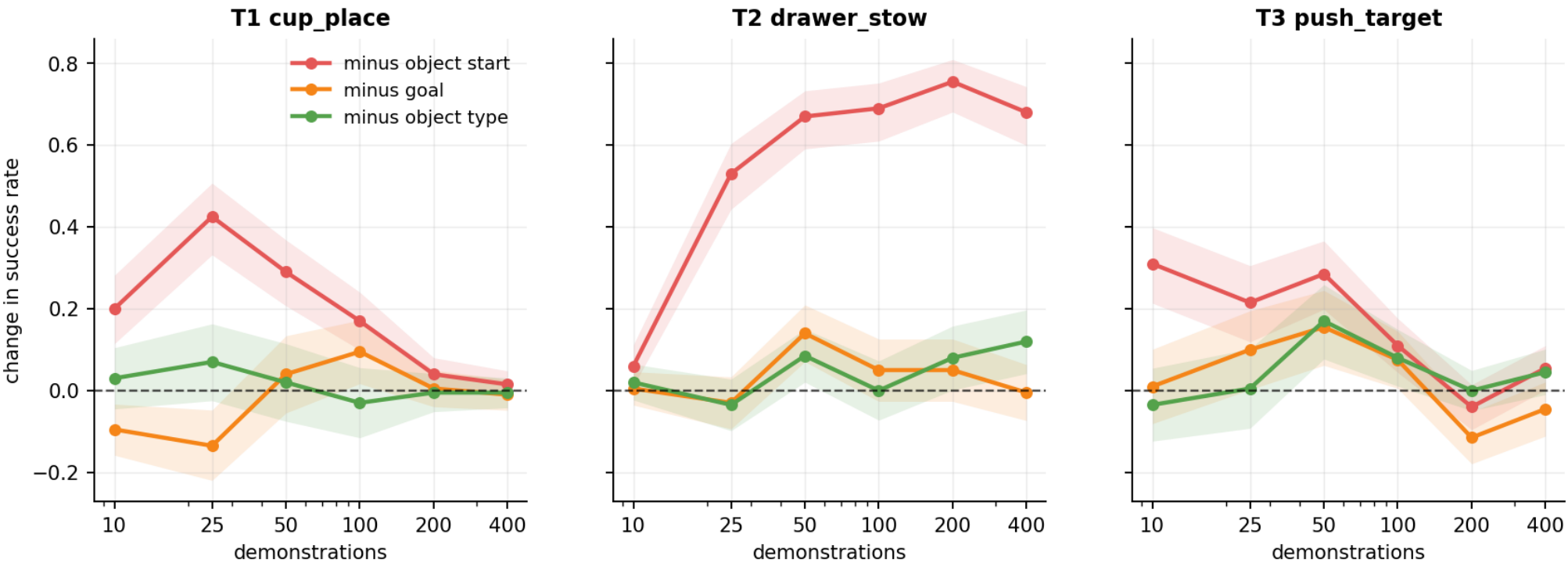
Four setups per task. Every one is trained at six demonstration budgets (10, 25, 50, 100, 200, 400) and scored on 200 episodes, so  $4 \times 6 \times 3$  tasks = 72 policies and 14,400 episodes. Same training recipe everywhere, one seed, 80,000 steps.

| setup              | object start position | goal position | object type | name on disk |
|--------------------|-----------------------|---------------|-------------|--------------|
| all on             | varies                | varies        | 10 types    | L3b          |
| minus object start | FIXED                 | varies        | 10 types    | BC           |
| minus goal         | varies                | FIXED         | 10 types    | AC           |
| minus object type  | varies                | varies        | 1 type      | L2           |

The three tasks are T1 cup\_place (put a cup on a goal disk), T2 drawer\_stow (open a drawer and stow an object in it) and T3 push\_target (push a puck to a target). The change reported for a disturbance is always **success rate without it, minus success rate with all three on, at the same demonstration budget**. Positive means the simpler setup did better.

# What dropping each disturbance does

## What dropping one disturbance does, against keeping all three



Change in success rate against keeping all three disturbances. Bands are 95 % intervals. Above the dashed line the simpler setup did better.

### T1 cup\_place

| demonstrations | minus object start                          | minus goal                                  | minus object type                           |
|----------------|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| 10             | +0.200 [+0.115, +0.282]                     | -0.095 [-0.158, -0.033]                     | +0.030 [-0.045, +0.105] (not clear of zero) |
| 25             | +0.425 [+0.332, +0.507]                     | -0.135 [-0.220, -0.048]                     | +0.070 [-0.025, +0.163] (not clear of zero) |
| 50             | +0.290 [+0.207, +0.368]                     | +0.040 [-0.055, +0.134] (not clear of zero) | +0.020 [-0.075, +0.115] (not clear of zero) |
| 100            | +0.170 [+0.099, +0.240]                     | +0.095 [+0.017, +0.172]                     | -0.030 [-0.115, +0.056] (not clear of zero) |
| 200            | +0.040 [+0.006, +0.080]                     | +0.005 [-0.039, +0.050] (not clear of zero) | -0.005 [-0.052, +0.041] (not clear of zero) |
| 400            | +0.015 [-0.014, +0.048] (not clear of zero) | -0.010 [-0.048, +0.027] (not clear of zero) | -0.005 [-0.042, +0.031] (not clear of zero) |

## T2 drawer\_stow

| demonstrations | minus object start      | minus goal                                  | minus object type                           |
|----------------|-------------------------|---------------------------------------------|---------------------------------------------|
| 10             | +0.060 [+0.011, +0.112] | +0.005 [-0.036, +0.046] (not clear of zero) | +0.020 [-0.023, +0.065] (not clear of zero) |
| 25             | +0.530 [+0.443, +0.604] | -0.030 [-0.094, +0.033] (not clear of zero) | -0.035 [-0.098, +0.028] (not clear of zero) |
| 50             | +0.670 [+0.591, +0.733] | +0.140 [+0.071, +0.209]                     | +0.085 [+0.021, +0.150]                     |
| 100            | +0.690 [+0.610, +0.752] | +0.050 [-0.026, +0.126] (not clear of zero) | +0.000 [-0.072, +0.072] (not clear of zero) |
| 200            | +0.755 [+0.681, +0.809] | +0.050 [-0.026, +0.126] (not clear of zero) | +0.080 [+0.001, +0.158]                     |
| 400            | +0.680 [+0.599, +0.743] | -0.005 [-0.073, +0.063] (not clear of zero) | +0.120 [+0.042, +0.197]                     |

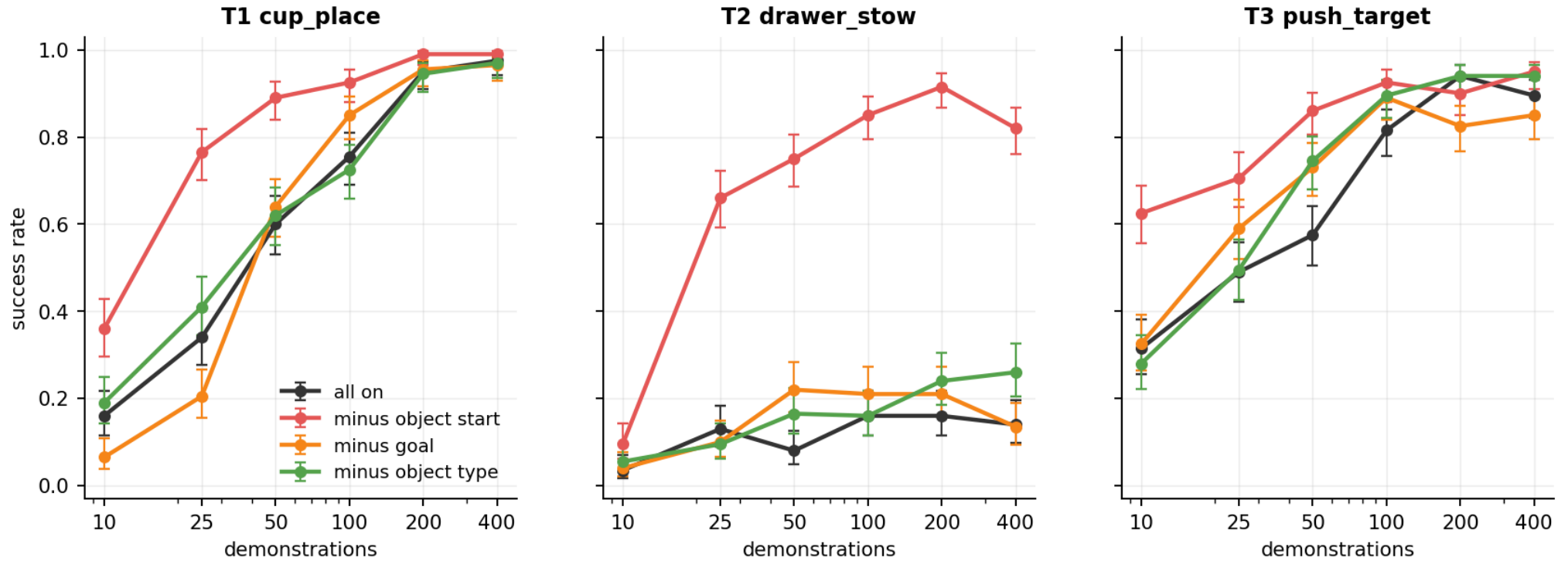
## T3 push\_target

| demonstrations | minus object start                          | minus goal                                  | minus object type                           |
|----------------|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| 10             | +0.310 [+0.214, +0.398]                     | +0.010 [-0.081, +0.101] (not clear of zero) | -0.035 [-0.124, +0.054] (not clear of zero) |
| 25             | +0.215 [+0.119, +0.305]                     | +0.100 [+0.002, +0.195]                     | +0.005 [-0.092, +0.102] (not clear of zero) |
| 50             | +0.285 [+0.199, +0.366]                     | +0.155 [+0.062, +0.245]                     | +0.170 [+0.077, +0.259]                     |
| 100            | +0.110 [+0.044, +0.176]                     | +0.075 [+0.005, +0.145]                     | +0.080 [+0.011, +0.149]                     |
| 200            | -0.040 [-0.096, +0.014] (not clear of zero) | -0.115 [-0.179, -0.052]                     | +0.000 [-0.049, +0.049] (not clear of zero) |
| 400            | +0.055 [+0.002, +0.110]                     | -0.045 [-0.111, +0.021] (not clear of zero) | +0.045 [-0.010, +0.101] (not clear of zero) |

Read each budget on its own line. The same disturbance can matter at 25 demonstrations and not at 400.

## The success rates behind those changes

Success rate of each setup



Success rate of each setup. Error bars are 95 % intervals on 200 episodes.

### T1 cup\_place

| demonstrations | all on          | minus object start | minus goal      | minus object type |
|----------------|-----------------|--------------------|-----------------|-------------------|
| 10             | 0.160 (32/200)  | 0.360 (72/200)     | 0.065 (13/200)  | 0.190 (38/200)    |
| 25             | 0.340 (68/200)  | 0.765 (153/200)    | 0.205 (41/200)  | 0.410 (82/200)    |
| 50             | 0.600 (120/200) | 0.890 (178/200)    | 0.640 (128/200) | 0.620 (124/200)   |
| 100            | 0.755 (151/200) | 0.925 (185/200)    | 0.850 (170/200) | 0.725 (145/200)   |
| 200            | 0.950 (190/200) | 0.990 (198/200)    | 0.955 (191/200) | 0.945 (189/200)   |
| 400            | 0.975 (195/200) | 0.990 (198/200)    | 0.965 (193/200) | 0.970 (194/200)   |

## T2 drawer\_stow

| demonstrations | all on         | minus object start | minus goal     | minus object type |
|----------------|----------------|--------------------|----------------|-------------------|
| 10             | 0.035 (7/200)  | 0.095 (19/200)     | 0.040 (8/200)  | 0.055 (11/200)    |
| 25             | 0.130 (26/200) | 0.660 (132/200)    | 0.100 (20/200) | 0.095 (19/200)    |
| 50             | 0.080 (16/200) | 0.750 (150/200)    | 0.220 (44/200) | 0.165 (33/200)    |
| 100            | 0.160 (32/200) | 0.850 (170/200)    | 0.210 (42/200) | 0.160 (32/200)    |
| 200            | 0.160 (32/200) | 0.915 (183/200)    | 0.210 (42/200) | 0.240 (48/200)    |
| 400            | 0.140 (28/200) | 0.820 (164/200)    | 0.135 (27/200) | 0.260 (52/200)    |

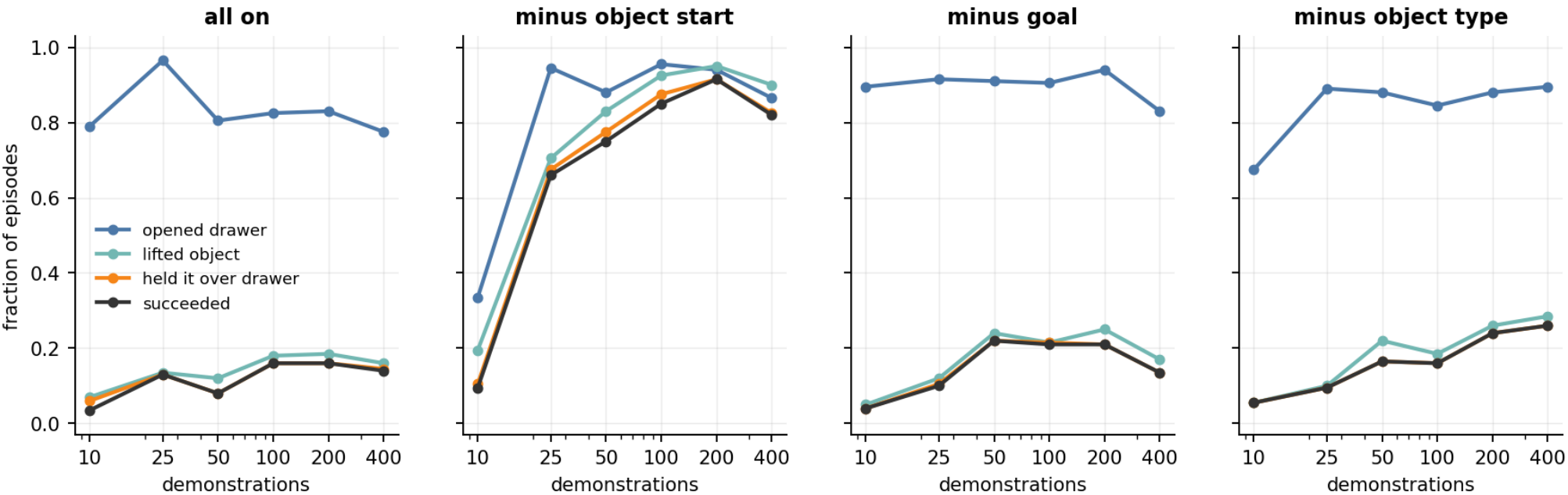
## T3 push\_target

| demonstrations | all on          | minus object start | minus goal      | minus object type |
|----------------|-----------------|--------------------|-----------------|-------------------|
| 10             | 0.315 (63/200)  | 0.625 (125/200)    | 0.325 (65/200)  | 0.280 (56/200)    |
| 25             | 0.490 (98/200)  | 0.705 (141/200)    | 0.590 (118/200) | 0.495 (99/200)    |
| 50             | 0.575 (115/200) | 0.860 (172/200)    | 0.730 (146/200) | 0.745 (149/200)   |
| 100            | 0.815 (163/200) | 0.925 (185/200)    | 0.890 (178/200) | 0.895 (179/200)   |
| 200            | 0.940 (188/200) | 0.900 (180/200)    | 0.825 (165/200) | 0.940 (188/200)   |
| 400            | 0.895 (179/200) | 0.950 (190/200)    | 0.850 (170/200) | 0.940 (188/200)   |

## Where T2 policies got stuck

T2 is the only task that records intermediate progress, so a failure can be placed rather than just counted. The milestones do not nest: a policy can lift the object without having opened the drawer far enough.

T2 drawer\_stow: how far each setup got



Fraction of 200 episodes reaching each milestone, per setup.

| setup              | demonstrations | opened drawer | lifted object | held over drawer | succeeded |
|--------------------|----------------|---------------|---------------|------------------|-----------|
| all on             | 10             | 0.790         | 0.070         | 0.060            | 0.035     |
| all on             | 25             | 0.965         | 0.135         | 0.130            | 0.130     |
| all on             | 50             | 0.805         | 0.120         | 0.080            | 0.080     |
| all on             | 100            | 0.825         | 0.180         | 0.160            | 0.160     |
| all on             | 200            | 0.830         | 0.185         | 0.160            | 0.160     |
| all on             | 400            | 0.775         | 0.160         | 0.145            | 0.140     |
| minus object start | 10             | 0.335         | 0.195         | 0.105            | 0.095     |
| minus object start | 25             | 0.945         | 0.705         | 0.675            | 0.660     |
| minus object start | 50             | 0.880         | 0.830         | 0.775            | 0.750     |

| setup              | demonstrations | opened drawer | lifted object | held over drawer | succeeded |
|--------------------|----------------|---------------|---------------|------------------|-----------|
| minus object start | 100            | 0.955         | 0.925         | 0.875            | 0.850     |
| minus object start | 200            | 0.940         | 0.950         | 0.915            | 0.915     |
| minus object start | 400            | 0.865         | 0.900         | 0.825            | 0.820     |
| minus goal         | 10             | 0.895         | 0.050         | 0.040            | 0.040     |
| minus goal         | 25             | 0.915         | 0.120         | 0.105            | 0.100     |
| minus goal         | 50             | 0.910         | 0.240         | 0.220            | 0.220     |
| minus goal         | 100            | 0.905         | 0.215         | 0.215            | 0.210     |
| minus goal         | 200            | 0.940         | 0.250         | 0.210            | 0.210     |
| minus goal         | 400            | 0.830         | 0.170         | 0.135            | 0.135     |
| minus object type  | 10             | 0.675         | 0.055         | 0.055            | 0.055     |
| minus object type  | 25             | 0.890         | 0.100         | 0.095            | 0.095     |
| minus object type  | 50             | 0.880         | 0.220         | 0.165            | 0.165     |
| minus object type  | 100            | 0.845         | 0.185         | 0.160            | 0.160     |
| minus object type  | 200            | 0.880         | 0.260         | 0.240            | 0.240     |
| minus object type  | 400            | 0.895         | 0.285         | 0.260            | 0.260     |

## What could be wrong with these numbers

| # | issue                                                                                        | what was measured                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>The two "minus" setups that need new demonstrations were generated on different GPUs.</b> | Demonstrations for minus-object-start and minus-goal came off A100s; the other two setups' demonstrations came off an RTX 4090. A control policy trained on A100-generated data scored 0.750 where its 4090-generated twin scored 0.860 (difference -0.110, 95 % CI -0.208 to -0.012). That control changed the GPU and the random seed together, so it bounds the effect rather than pinning it down. Any change in this report smaller than about 0.11 sits inside that boundary. |
| 2 | <b>One training seed.</b>                                                                    | Every policy is seed 0. The intervals here cover episode sampling only; they contain no training-run variance.                                                                                                                                                                                                                                                                                                                                                                      |
| 3 | <b>Repeating an evaluation does not give the identical episodes back.</b>                    | Re-running one 20-episode slice with the same seed and checkpoint reproduced the slice total but flipped two individual episodes. Expect about +/-1 success per 20 episodes of jitter.                                                                                                                                                                                                                                                                                              |
| 4 | <b>"Minus object type" is not a newly generated setup.</b>                                   | Removing the object axis from the full set gives exactly the existing single-object setup, verified identical over 300 matched resets. It is re-used rather than regenerated, which is also why it is the only comparison whose two sides see the same starting positions.                                                                                                                                                                                                          |
| 5 | <b>Do not average a change across demonstration budgets.</b>                                 | The six budgets are six different policies. A test for a shared success rate is rejected in all 18 setup-task pairs, so an average over budgets describes the mix of budgets chosen, not the setup.                                                                                                                                                                                                                                                                                 |

## What would fix them

| # | next step                                                                                                                           |
|---|-------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Re-run the GPU control with the demonstration seed held fixed, so the GPU is the only thing that changes.                           |
| 2 | Train more than one seed on whichever comparisons matter.                                                                           |
| 3 | Record the evaluator version in the results ledger, and which git revision each run used.                                           |
| 4 | Measure whether a second warm-up batch moves the scored success rate.                                                               |
| 5 | Guard the pooling script against mixing evaluator versions, and assert the termination-term ordering the success signal depends on. |

Every number here comes from `src/cog/analysis/loo.py`, over the 108 pooled result files in `results/`. The long edition, with all six arms of the sweep, the per-slice dispersion, timing distributions and the full flaw list, is `paper/loo_report/loo_report_full.pdf`. The tables are also plain CSV in `paper/loo_report/tables/`.